Attention-Enhanced ResNet34 with Transfer Learning for Brain Tumor MRI Classification

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Abstract—Accurate classification of brain tumors from Magnetic Resonance Imaging (MRI) is critical for early intervention and treatment planning. This study proposes an enhanced ResNet34 framework (TResNet34_CBAM) that integrates Transfer Learning with the Convolutional Block Attention Module (CBAM). By employing balanced data augmentation and a dual-attention mechanism, the model addresses the challenges of medical data scarcity and complex pathological feature extraction. Experimental results on a dataset of 7,023 images demonstrate that our proposed model achieves an accuracy of 98.06% and an F1-score of 0.98, significantly outperforming baseline ResNet34 and EfficientNet architectures.

Index Terms—Brain Tumor Classification, ResNet34, CBAM, Transfer Learning, MRI, Computer-Aided Diagnosis (CAD).

I. INTRODUCTION

The diagnosis of brain tumors remains one of the most challenging tasks in modern oncology due to the high complexity of the human neurological system. Brain tumors, if not detected and localized in their early stages, can lead to severe cognitive impairment, physical disability, and high mortality rates. Magnetic Resonance Imaging (MRI) has emerged as the definitive non-invasive diagnostic tool, offering exceptional soft-tissue contrast that allows clinicians to visualize internal structures without ionizing radiation. However, the rapidly increasing volume of medical imaging data has placed an overwhelming burden on radiologists. Manual interpretation is not only time-consuming but also inherently susceptible to human fatigue and inter-observer variability, which may lead to delays in critical treatment planning. Consequently, there is an urgent clinical need for automated, high-precision computer-aided diagnosis (CAD) systems that can assist medical professionals in providing rapid and reliable second opinions.

To address these challenges, deep learning—specifically Convolutional Neural Networks (CNNs)—has been extensively explored for medical image classification. While large-scale architectures often achieve high performance, they frequently encounter issues such as over-fitting when applied to specialized medical datasets, which are often smaller and more imbalanced than general-purpose datasets like ImageNet. Conversely, lighter models like ResNet34 [1] offer better computational efficiency and lower risk of overfitting but may lack the representational capacity to capture the subtle pathological nuances required for multi-class tumor differentiation.

This research proposes an optimized framework, TResNet34_CBAM, which integrates Transfer Learning with the Convolutional Block Attention Module (CBAM). By focusing on both "what" (channel-wise) and "where" (spatial-wise) features, our approach aims to enhance the sensitivity of ResNet34 to tumor-specific regions, ultimately achieving a classification accuracy that exceeds clinical benchmarks while maintaining model efficiency.

II. RELATED WORK

A. Transfer Learning

Transfer learning [2] is a pivotal technique in deep learning, designed to leverage knowledge acquired from a large-scale source domain and apply it to a target domain. This approach addresses the challenge of data scarcity by utilizing pre-trained models, thereby eliminating the need to train deep networks from scratch. The primary advantage is the significant reduction in training time and computational resources, while simultaneously improving generalization on smaller medical datasets.

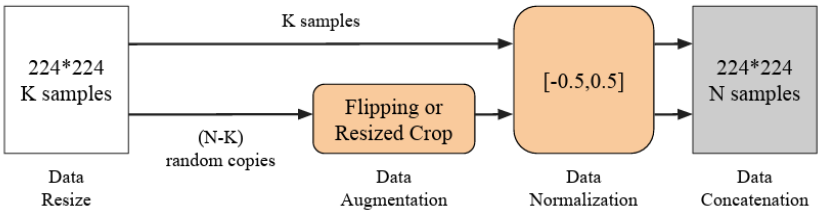

Figure 1. Data Preprocessing

The standard procedure involves utilizing the weights from a model trained on a massive dataset like ImageNet and fine-tuning all layers to recognize task-specific patterns in MRI scans. This enables the network to stabilize faster and achieve higher accuracy even with limited clinical data.

B. Data augmentation

Data augmentation is essential for preventing overfitting and improving the generalization ability of models in medical imaging, where labeled data is often scarce or imbalanced. Common techniques include geometric transformations such as horizontal and vertical flipping, as well as random resized cropping. These methods artificially expand the training set, allowing the model to become invariant to the orientation and scale of the tumor within the MRI slice. By balancing the number of samples across different tumor classes—such as glioma, meningioma, and pituitary tumors—augmentation helps stabilize the training process and ensures the classifier does not develop a bias toward the majority class.

C. Attention Mechanisms

Attention mechanisms are widely used to boost CNN performance by suppressing irrelevant information and focusing on salient regions. The Squeeze-and-Excitation (SE) Network introduced the concept of channel attention, which recalibrates the importance of different feature channels to emphasize informative ones. Building upon this, the Convolutional Block Attention Module (CBAM) [3] combines both channel and spatial attention mechanisms. Channel attention identifies "what" is important in the image, while spatial attention identifies "where" the most informative regions are located. By integrating CBAM into architectures like ResNet34, the model can more accurately pinpoint tumor locations while ignoring background noise and irrelevant anatomical structures.

III. PROPOSED METHOD

1. Data Preprocessing

The original dataset exhibits an imbalance across various tumor types. To mitigate this, we employed data augmentation techniques, as illustrated in **Figure 1**, including flipping and resized cropping until each class reached 1,600 training samples and 400 test samples. All images were resized to a standard 224*224 resolution and normalized to the range of [-0.5, 0.5] to ensure consistent input distribution.

*2. Model Architecture (shown in **Figure 2**)*

Our framework, TResNet34_CBAM, is designed to maximize the feature extraction capabilities of residual networks while adapting them to the specific nuances of medical imaging. To enhance the pre-trained ResNet34 backbone, there are four key modifications implemented:

1. *G2B Conversion*: A Conv2d(1,3) layer is added to convert one-channel grayscale MRI images into a three-channel format compatible with pre-trained weights.
2. Feature Extraction Backbone: We utilize a pre-trained ResNet34 to benefit from efficient feature reuse via transfer learning.
3. Attention Refinement: At the final stage of the backbone (Layer 4), we embed the CBAM module in the last BasicBlock to refine high-level features. The process occurs in two sequential steps—
    a. Channel Attention evaluates the importance of each feature map, allowing the network to decide "what" pathological features are most relevant. The refined map is generated by the formula:

$$X_c = \sigma(MLP(AvgPool(X_{in})) + MLP(MaxPool(X_{in})))$$

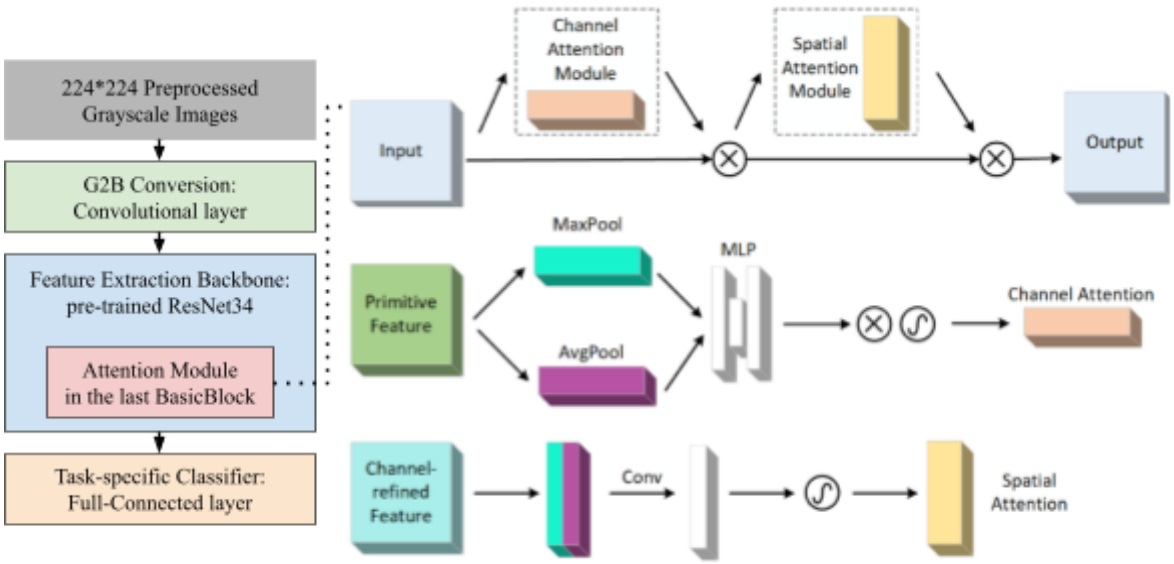

Figure 2. TResNet34_CBAM Structure

Where σ denotes the sigmoid function and MLP is a multilayer perceptron with a shared hidden layer. This mechanism helps the model focus on informative channels and suppress irrelevant background information.

b. Then, a *Spatial Attention* map is generated to highlight "where" the tumor-related pixels are located, effectively filtering out non-brain background noise. And this map can be expressed as:

$$X_s = \sigma(f^{7*7}([AvgPool(X_c); MaxPool(X_c)]))$$

Where f^{7*7} is a convolution operation with a filter size of 7*7. This allows the network to accurately pinpoint brain tumor locations, which is a significant improvement over standard CNN architectures.

4. *Four-class Classification*: The original fully connected layer is replaced with a task-specific classifier tailored for four-class brain tumor prediction.

IV. EXPERIMENTS

We conduct experiments on a composite dataset comprising 7,023 brain MRI images and 4 classes, which combine the Figshare, SARTAJ, and Br35H datasets [4]. Experiments were conducted in Google Colab using the Adam optimizer with a learning rate of 5e-5, cross-entropy loss function, a batch size of 8, and 5 training epochs. A 5-fold cross-validation with an 80:20 train-validation stratified split was employed for validation, and four indices were used for quantitative evaluation, including the accuracy, precision, recall, and F1-score.

1) Comparison to other Networks:

The performance of the proposed model TResNet34_CBAM was evaluated against several baseline architectures to verify its superiority in brain tumor classification. As shown in **Table 1**, the baseline ResNet34 established a solid performance with an accuracy of 91.38%. However, more complex and parameter-efficient models like EfficientNet B0 and EfficientNet B1 achieved significantly lower accuracies of 84.81% and 84.00%, respectively. This performance gap suggests that while EfficientNet is designed for efficiency, it may underfit the complex and subtle pathological patterns found in brain MRI scans. In contrast, our model reached an accuracy of 98.06%, outperforming the standard ResNet34 by 6.68 percentage points. All other metrics—precision, recall, and F1-score—consistently reached 0.98, demonstrating that the integration of transfer learning and attention mechanisms allows the model to achieve a more balanced and robust classification across all four tumor types.

Models	Accuracy (%)	Precision	Recall	F1-score
(Baseline) ResNet34	91.38	0.91	0.91	0.91
EfficientNet_B0	84.81	0.85	0.85	0.85
EfficientNet_B1	84.00	0.85	0.84	0.84
(Ours) TResNet34_CBAM	98.06	0.98	0.98	0.98

Table 1. Results of TResNet34_CBAM Compared with Other Models

TResNet34	Accuracy (%)	Precision	Recall	F1-score
None	96.92	0.97	0.97	0.97
ChAt	97.31	0.97	0.97	0.97
SpAt	97.88	0.98	0.97	0.97
CBAM	98.06	0.98	0.98	0.98

Table 2. Results of Ablation Experiments for Attention Mechanism

2) Ablation Study:

We investigate our model by examining the attention mechanism. As **Table 2** demonstrated, The baseline (None) achieved 96.92% accuracy, benefiting primarily from transfer learning and data augmentation. Upon introducing the channel-based attention (ChAt), the accuracy improved to 97.31% indicates that recalibrating channel-wise features helps the model emphasize informative tumor characteristics. The inclusion of spatial attention (SpAt) alone yielded a higher improvement to 97.88%, highlighting the importance of pinpointing the specific location of the tumor within the MRI slice. Ultimately, the combined CBAM module, which integrates both channel and spatial attention, achieved the peak accuracy of 98.06%. This confirms that the dual-attention mechanism effectively suppresses irrelevant background noise and enhances the model's focus on the most critical pathological regions.

V. CONCLUSION

This study introduced an Attention-Enhanced ResNet34 framework that integrates balanced data augmentation, transfer learning, and CBAM successfully. Our model reached 98.06% accuracy and 0.98 for all evaluation metrics, demonstrating that the addition of CBAM provides consistent improvements over baseline models. The proposed architecture is lightweight and efficient, making it suitable for real-world deployment on limited hardware while assisting clinicians in early and reliable tumor diagnosis.

VI. REFERENCES

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